HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and executed small-scale bioprocess technology experiments for vaccine manufacturing process characterization, adhering strictly to **safety and regulatory requirements**.
- Applied principles of **bioprocess engineering** and **biochemistry** to process development projects, contributing to the optimization of manufacturing workflows.
- Collaborated closely with **analytical scientists** on process development studies, integrating laboratory findings into broader project objectives.
- Contributed to team discussions, applying an **innovative mindset** to develop economical, efficient, and safe bioprocessing technologies.
- Executed **laboratory-based** components of experimental programs, gaining practical experience in bioprocessing and separation sciences.
- Troubleshot technical issues related to complex instrumentation and software within the biopharmaceutical manufacturing environment, planning and executing experiments independently.
- Managed production support for enterprise **Java/.NET applications** using **Splunk** and **Dynatrace**, ensuring SLA adherence and responding to P1/P2 incidents.

SKILLS

Biotechnology & Engineering – Bioprocessing, Separation Sciences, Bioprocess Engineering, Biochemical Engineering, Vaccine Manufacturing, Drug Development

Laboratory & Scientific Skills – Process Development, Experimental Design, Laboratory Execution, Instrumentation, Process Modelling, Biochemistry, Cell Biology, Microbial Fermentation, Genetics

Software & Tools – Power BI, Tableau, R/RStudio, Python (pandas, NumPy, scikit-learn), SQL, Java, Spring Boot, REST APIs, AWS, Docker, Linux, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace

Data Analysis & BI – Data Analytics, Statistical Analysis, Regression, Hypothesis Testing, Exploratory Data Analysis, A/B Testing, Time-Series Analysis, Business Intelligence, Data Modelling, DAX, Power Query

Project Management & Compliance – Agile/Scrum, Change Management, Risk Analysis, Regulatory Requirements, GDPR Awareness, Documentation

PROJECTS

Biopharmaceutical Process Optimization Study

- Designed and executed a series of experiments to characterize and improve a key separation process within a biopharmaceutical manufacturing workflow, focusing on yield and purity.
- Analyzed experimental data using **Python (pandas, NumPy)** and **R/RStudio**, applying statistical methods to identify optimal operating parameters and reduce process variability by **18%**.
- Documented experimental procedures, results, and recommendations in a comprehensive report, facilitating knowledge transfer to the wider process development team.

Enterprise Application Support & Automation

- Provided production support for critical enterprise applications, diagnosing and resolving complex incidents impacting service availability across multiple environments.
- Developed **Python scripts** to automate routine data validation and reporting tasks for financial operations dashboards, reducing manual effort by approximately **10 hours per week**.
- Managed change control processes for system updates and configuration changes, coordinating User Acceptance Testing (UAT) and post-go-live support across interconnected systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)